

Analyzing the Performance of Generalized Hill Climbing Algorithms

Abstract

Generalized Hill Climbing (GHC) algorithms provide a unified framework for describing a broad class of local search and metaheuristic optimization techniques used to solve hard discrete optimization problems. This paper presents a comprehensive analysis of GHC algorithms, focusing on their convergence properties, performance evaluation, and comparison with random restart local search. A key contribution is the discussion of the finite global visit probability as a practical performance metric that captures finite-time behavior more effectively than asymptotic convergence results. Theoretical convergence conditions are reviewed, with simulated annealing examined as a special case of the GHC framework. The paper concludes that while simulated annealing offers strong theoretical guarantees, random restart strategies and hybrid approaches often achieve superior performance in realistic computational settings.

1. Introduction

Discrete optimization problems appear in many real-world applications such as scheduling, network design, logistics, and machine learning. These problems often have large and complex search spaces, making exact solution methods computationally infeasible. As a result, heuristic and metaheuristic algorithms are widely used to obtain high-quality solutions within reasonable time limits.

Generalized Hill Climbing (GHC) algorithms form a powerful class of such methods. They extend traditional hill climbing by incorporating probabilistic acceptance mechanisms that allow the search process to escape local optima. Well-known algorithms such as simulated annealing, threshold accepting, and the noising method can all be expressed within the GHC framework. This unification enables a consistent theoretical analysis of their convergence behavior and performance characteristics.

2. Literature Review

Early work on local search and combinatorial optimization highlighted the difficulty of escaping local optima using purely deterministic methods. Simulated annealing, inspired by statistical physics, introduced stochastic acceptance rules and was shown to converge to global optima under carefully controlled cooling schedules. Hajek (1988) provided necessary and sufficient conditions for the convergence of simulated annealing.

Subsequent research explored alternative metaheuristics, including threshold accepting, tabu search, and random restart strategies. While many studies focused on asymptotic convergence, others argued that finite-time performance is more relevant in practice. The concept of random restart local search emerged as a simple yet effective approach, particularly for landscapes with many local optima.

3. Generalized Hill Climbing Framework

A GHC algorithm operates on a finite solution space equipped with an objective function and a neighborhood structure. At each iteration, a neighboring solution is randomly generated and accepted or rejected based on a hill-climbing random variable. This probabilistic rule allows occasional uphill moves, enabling exploration beyond local optima.

The solution space can be partitioned into global optima, local optima, and non-optimal solutions. The neighborhood function defines the landscape structure and plays a critical role in determining algorithm performance. Reachability of the solution space is a key assumption for convergence analysis.

4. Finite Global Visit Probability

The finite global visit probability measures the likelihood that a GHC algorithm visits at least one global optimum within a finite number of macro iterations. A macro iteration represents a transition between local or global optima, possibly through non-optimal states.

This metric provides a practical way to evaluate algorithm performance under limited computational budgets. Unlike asymptotic convergence results, it reflects the probability of success within a realistic execution horizon.

5. Convergence Analysis

The convergence of GHC algorithms can be characterized using necessary and sufficient conditions based on the probability of visiting global optima. In particular, convergence in probability requires that the cumulative probability of first reaching a global optimum diverges and that the algorithm increasingly remains at global optima once reached.

Simulated annealing is shown to satisfy these conditions when its cooling schedule decreases slowly enough. This establishes simulated annealing as a special case of the GHC framework and demonstrates the generality of the convergence theory.

6. Comparison with Random Restart Local Search

Random restart local search repeatedly applies deterministic local search from randomly selected initial solutions. Each restart corresponds to a macro iteration. This approach is particularly effective when the probability of falling into non-global local optima is high.

Theoretical results indicate that random restart local search eventually outperforms non-convergent GHC algorithms. For convergent GHC algorithms, relative performance depends on neighborhood structure, cooling schedules, and the distribution of attraction basins.

7. Discussion

The analysis highlights the importance of designing effective neighborhood functions and parameter settings. Algorithms with strong asymptotic guarantees may still perform poorly in practice if convergence is too slow. Hybrid strategies that combine random restarts with stochastic local search often provide the best balance between exploration and exploitation.

8. Conclusion

This paper examined the performance of Generalized Hill Climbing algorithms through a unified theoretical framework. By introducing the finite global visit probability, the analysis emphasized finite-time behavior and practical effectiveness. While simulated annealing remains a foundational metaheuristic, random restart and hybrid approaches are often more suitable for real-world optimization problems.

References

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